SMART RESTAURANT MANAGEMENT SYSTEM

Project Overview

Vision, scope, journey and delivery plan

DOCUMENT ID	SRMS-01
VERSION	1.0
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STATUS	Draft for review
PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	Explains what we are building, why, for whom, and how the project is phased. Start here.

1. Introduction

The **Smart Restaurant Management System** lets guests scan a QR code on their table, browse the menu, place orders and pay from their own phone, while giving the restaurant manager a **live view of the whole floor**: which tables are empty, which need cleaning, and which need attention right now.

The idea was derived from a short video on AI-assisted restaurant floor management, which shows camera-based table status tracking (free / occupied, guest counts, dwell timers, wait times and a live floor map). This project delivers the same outcome in two phases: first an **event-driven system** (QR, orders, payments and staff actions drive table status), then an **AI camera detection layer** that feeds the same status engine.

2. Problem Statement

Problem today	Impact
Managers cannot see table status without walking the floor	Slow seating, idle empty tables, unhappy waiting guests
Guests wait for a waiter to take orders, bring the bill and process payment	Longer table time, lower turnover, lost tips and sales
Cleaning is triggered by chance, not by a signal	Dirty tables stay unused; inconsistent hygiene
No alert when a table is ignored or an order is delayed	Poor service recovery, bad reviews
Order errors from verbal/paper tickets	Wastage, refunds, kitchen confusion
No data on turnover, dwell time or peak hours	Guess-based staffing and decisions

3. Vision and Objectives

Vision: every table in the restaurant is self-service for the guest and fully visible to the manager in real time.

- Let guests order and pay from a table QR code without installing an app.
- Give the manager a real-time floor map with four clear states: Empty, Occupied, Needs Cleaning, Needs Attention.
- Route orders instantly to the kitchen and track each item to the table.
- Create automatic tasks for waiters and cleaners and alert on delays.
- Support UPI, cards and cash with correct GST billing.
- Provide reports on turnover, dwell time, peak hours and best-selling items.
- Prepare the architecture so AI camera detection plugs in later without redesign.

4. Scope

In scope: Phase 1 (event-driven)	In scope: Phase 2 (AI camera)	Out of scope (for now)
QR table sessions Digital menu and cart Order placement and kitchen display Online payments and cash recording Live floor map and table state engine Alerts: call waiter, delays, bill request Cleaning task queue Staff roles and admin panel Basic reports	Camera zones mapped to tables Person and object detection Auto detect: seated, left, table cleared Dwell and wait-time estimates Entrance queue counting Confidence-based suggestions to staff Privacy controls (no face storage)	Inventory and procurement Accounting software integrations Delivery aggregator integration Loyalty program Multi-country tax rules Native mobile apps (PWA first)

5. Stakeholders

Stakeholder	Interest
Restaurant owner	Revenue growth, multi-outlet visibility, cost control

Stakeholder	Interest
Manager	Live floor control, alerts, staff coordination, reports
Waiters	Clear tasks, fewer interruptions, less order taking
Kitchen staff	Accurate, prioritised, timed tickets
Cleaning staff	Know exactly which table to clean and when
Guests	Fast, contactless ordering and payment; less waiting

6. Customer Journey (End to End)

1. Guest sits at a table and scans the QR code. The code carries a secure table token.
2. The system opens (or joins) a **table session** and marks the table **Occupied**. The manager sees it turn red.
3. Guest browses the menu, customises items and adds them to the cart. Several guests at one table share the same cart.
4. Guest places the order. The kitchen display receives it. Waiter is notified for confirmation if configured.
5. Item statuses move Accepted, Preparing, Ready, Served. The guest sees live progress.
6. Guest can tap **Call Waiter** or **Request Bill**. This raises a **Needs Attention** alert on the manager map.
7. Guest pays by UPI/card, or chooses cash. Payment success closes the session.
8. Table moves to **Needs Cleaning**. A cleaning task is assigned. When the cleaner taps Done, the table becomes **Empty**.

7. What the Video Taught Us

Observed in video	Our Phase 1 equivalent	Phase 2 (AI) equivalent
Green = free table, red = occupied	Status from QR scan, order and payment events	Person detected in table zone
Table label with guest count and timer	Guest count from session; timer from session start	Count people in zone
Occupancy panel (tables, guests, capacity %)	Calculated from live table states	Same panel, AI-verified
Wipe-down turns table to FREE	Cleaner taps Done	Detect cleaner activity and clear table
Wait time and entrance queue	Waitlist with manual add; wait timer	Count people at entrance
Long-stay timer (e.g. 1 hour 15 min)	Session duration alerts	Dwell-time analytics
Detect customer and staff need	Call waiter, order delay, inactivity alerts	Detect raised hand or ignored table

8. Delivery Phases

Phase	Goal	Outcome
Phase 1	Event-driven core	Working QR ordering, payments, manager floor map, alerts, cleaning tasks
Phase 2	AI camera detection	Cameras confirm or auto-update table status; queue and dwell analytics
Phase 3	Scale and growth	Multi-outlet, reservations, loyalty, deeper analytics

9. Success Metrics (KPIs)

KPI	Target (first 3 months of use)
Share of orders placed via QR	At least 70%

KPI	Target (first 3 months of use)
Average table turnover time	Reduce by 15%
Time from guest leaving to table ready	Under 5 minutes
Alert acknowledge time (call waiter)	Under 90 seconds
Order accuracy (no manual re-entry)	At least 98%
Payment success rate (online)	At least 97%
Floor map accuracy vs reality (Phase 1)	At least 95%

10. Assumptions and Constraints

- Guests have a smartphone with a camera and mobile data or restaurant Wi-Fi.
- The restaurant has reliable internet; kitchen and manager screens are tablets, PCs or phones.
- Primary market is India: INR currency, UPI payments, GST invoicing, Indian data-protection norms.
- Staff will tap status buttons (cleaned, served). Phase 2 reduces this effort but does not remove manual override.
- Phase 1 is a responsive web app (PWA). Native apps are optional later.

11. Glossary

Term	Meaning
Table session	The period from first scan or seating until payment is completed and the table is released
Needs Attention	An overlay flag raised when a table has an open alert such as a waiter call or delayed order
KDS	Kitchen Display System, the screen where the kitchen sees tickets
PWA	Progressive Web App; a website that behaves like an app, no store install needed
Dwell time	How long guests occupy a table
State engine	The backend component that is the single source of truth for table status